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Financial markets co-movement predictive modeling: the DWT approach

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Abstract

Trends in capital markets are difficult to predict due to the complex linkages. Recent research has shifted from predicting single market trends to predicting multiple market co-movements. Based on the WTI oil price, Douglas stock index and Shenzhen composition index, we use wavelet transformation to extract the stable domains and identify the co-movement directions. Next, we construct a spectrum plot for co-movement prediction. Experimental results show that a discrete wavelet transform is effective in predicting financial market co-movements within the three markets.

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Keywords: Financial market co-movement; Discrete wavelet transform; Spectrum plot

1. Introduction

Co-movement has an increasingly important role in modern finance, which is not only influenced by the global economic development, but also influenced by the investment strategies within financial institutions. These investments further influence economic growth. Related research can be found in the work of Farago and Hjalmarsen (2018)¹ and Kiviahio, Nikkinen, Piljak, and Rothovius (2014)². They argued that investors and officers should pay more attention to financial correlation. Today, more and more global financial markets are opened with limited boundaries; thus correlation among different markets is getting tighter. Pradhananga (2016)³ and Han (2019)⁴ tracked the trend of different financial markets. Gangopadhyay, Nishimura, and Pal (2016)⁵ believe that co-

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movement is one of the results of globalization and it has been accelerated by the development of information technology. In recent research, the phenomenon is defined as financial market co-movement. Trends in the energy market strongly indicate the performance of the financial market. Therefore, investigating the co-movement between the two markets will shed light on how external shocks influence the financial market, and how the economy changes as a result of Yin, Yang, & Su (2017)⁶.

In the past few decades, empirical studies of stock markets have established the efficient market hypothesis. Stock predictions are the focus of institution and researchers; they seek an effective model to predict the price in single market or to see if there is a general law to predict lift or not. Many studies have employed methods from econometrics, time series analysis, machine learning, and wavelet analysis. Under the efficient market hypothesis, however, these methods are effective in a single financial market with the price following the stochastic process. Recently, researchers have turned the focus to finding the cause of financial market volatility. There are two main factors that can be used to explain the phenomena to a degree: one is seen as the internal factor—cash flow; and the other is the outer factor—correlation. However, many studies on financial market co-movements violate the efficient market assumption to a certain extent; besides, co-movement needs to adopt different approaches that are not auto regression, so time series does not work. One approach is to analyze financial market co-movements under external shocks, which can inform the signal of financial market forecasting. Furthermore, the Chinese stock market is different from the free market; it does not follow the effective market hypothesis. What is more, although it has more and more impact on global finance, research on financial markets co-movement with China's A stock market is still scarce.

2. Literature Review

Usually, scholars' initial research on the phenomenon of co-movement started from the connection between markets, so there has been some discussion on how to describe the linkage between separated markets, seen in Graham and Nikkinen (2011)⁷, they examined the short term and long term co-movement of European international stock markets. Pastpipatkul, Yamaka, and Sriboonchitta (2016)⁸ explored the co-movement between gold market with Indonesian, Philippine, and Thailand stock markets. Saâdaoui, Naifar, and Aldohaiman (2017)⁹ used wavelet-assisted cross-spectral model to discuss dynamic correlation between the conventional and Islamic stock markets. Sahabuddin, Muhammad, HjYahya, Shah, and Rahman (2018)¹⁰ used time series to investigate the co-movement relationship between compliant stock indexes and Bursa Malaysia sector stocks indexes.

An, Gao, Fang, Ding, and Zhong (2014)¹¹ explored the transformation characteristics of co-movement between crude oil futures and spot prices. These changes illustrate some reason for the correlation among different markets. Ftiti, Guesmi, and Abid (2016)¹² studied the correlation between oil and stock markets in G7 countries. Ghosh and Kanjilal (2016)¹³ explored a multivariate framework which can investigate nonlinear co-movement between the international crude oil price and the Indian stock market. Huang, Huang, An, and Gao (2016)¹⁴ turned to explore the correlation of crude oil, gold and the Shanghai Composite index in a joint time-frequency domain, and then they proposed a framework to analyze the co-movement within the time domain and frequency domain. Bunn, Chevallier, Le Pen, and Sevi (2017)¹⁵ investigated whether common factors had a strong impact on US indices. Lately, researchers have put more efforts on direct relationship analyses; Maghyreh, Awartani, and Abdoh (2019)¹⁶ discussed oil prices and clean energy stock prices, which may be a reasonable comparison for co-movement in different financial markets.

3. Wavelet analysis

We collected the daily closing prices of the WTI, the DJI and the Shenzhen composition index (SZSE) from January 2009 to December 2018. The trend of these indexes can be seen in Figure 1. From Figure 1, we can see that the DJI index had a clear wave of shocks during 2013. In order to make the results more robust, we selected data from January 2014 to January 2018 for analysis.

The main statistical parameters and statistical results of the above indices are shown in Table 1. From Table 1, it can be inferred that a predictive analysis framework based on the efficient market hypothesis has a poor analytical effect in these markets⁴.

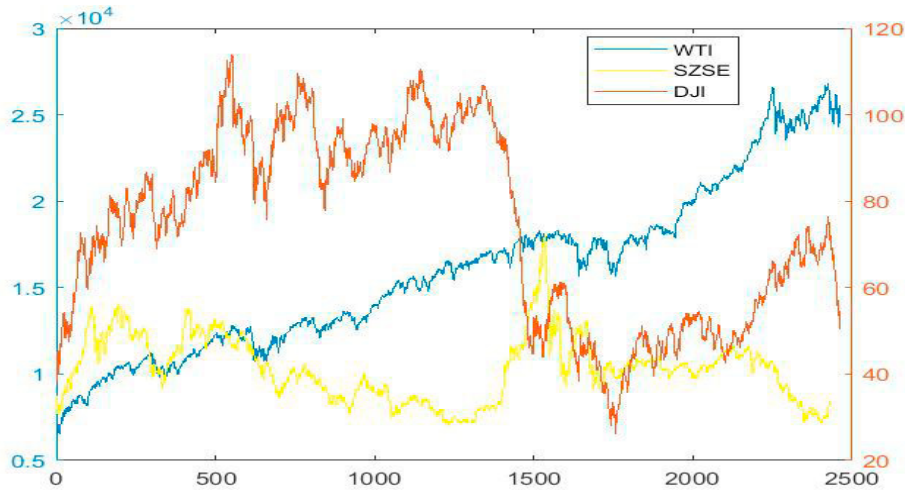


Fig. 1. Trends of the three indexes

Table 1. Descriptive Statistics and Test Results of Three Indexes

	WTI	DJI	SZSE
Minimum	1844	12118.57	6959.25
Maximum	7507	21005.71	18211.76
Mean	4663.74	16214.55	8055.73
Median	4992.50	16483.47	8437.78
Variance	1654774.20	4367662.34	1650153.58
Std	1286.38	2089.90	1284.58
Skewness	-0.351	-0.211	-0.105
Kurtosis	-0.956	-0.785	-1.535
Jarque-Bera	83.8115***	42.7251***	130.2585
Shapiro Wilk	0.9366***	0.9505***	0.9009***
H-G Test	0.0624***	0.0481***	0.0985***
Terasvirta's Test	9.3062***	6.8614**	9.15**
White's Test	8.0968***	5.6247*	7.3464**

*** Significant at a 1% level of significance, ** Significant at a 5% level of significance, * Significant at a 10% level of significance.

Wavelet analysis is a powerful quantitative modeling tool, which enables carrying out multilevel analysis to non-stationary time series. It is widely used in sequence decomposition and signal processing¹⁷. We used these frameworks to accomplish the research endeavors. Discrete wavelet transformation generates father and mother wavelets in a scale-wise manner by translating and dilating the original function¹⁷, it can be expressed as equation (1).

$$\psi_{a,b}(x) = |a|^{-\frac{1}{2}} \psi\left(\frac{x-b}{a}\right) \quad b \in R, a \in R^+ \quad (1)$$

The wavelet coherence coefficient ranges from 0 to 1. And if the value is close to 0, it indicates a weak dependence, and vice versa. The wavelet coherence is calculated to as follows¹⁸:

$$\theta_{xy}(s) = \tan^{-1} \left(\frac{\hat{h}(s^{-1}W_s^{ab}(s))}{\hat{\lambda}(s^{-1}W_s^{ab}(s))} \right) \quad (2)$$

where the imaginary and real components of the wavelets are represented by $\hat{h}$ and $\hat{\lambda}$.

Cross-wavelet plot can give a detailed explanation. Arrows in the plot indicate the phase relationships of the coherence phase. If the arrows are directed toward the right side, an inference can be drawn that there is positive correlation and the two time-series are in phase (they have the same frequencies); whereas, left facing arrows signify a negative correlation and a series that is out of phase. For our case, downward directed arrows signify that the second series lead the first by half-cycle, while upward directed arrows signify that the first series lead the second by half-cycle.

Figure 2 represents the cross-wavelet transformation between two time-series. Looking at the vertical axis, the dynamic relationship between any two financial series appear to be been more affected by medium to long term changes than by short term shocks. The interplay of phases is homogeneous across the scales, because the arrows point in all directions. So there is no general conclusion of the co-movement between these three markets.

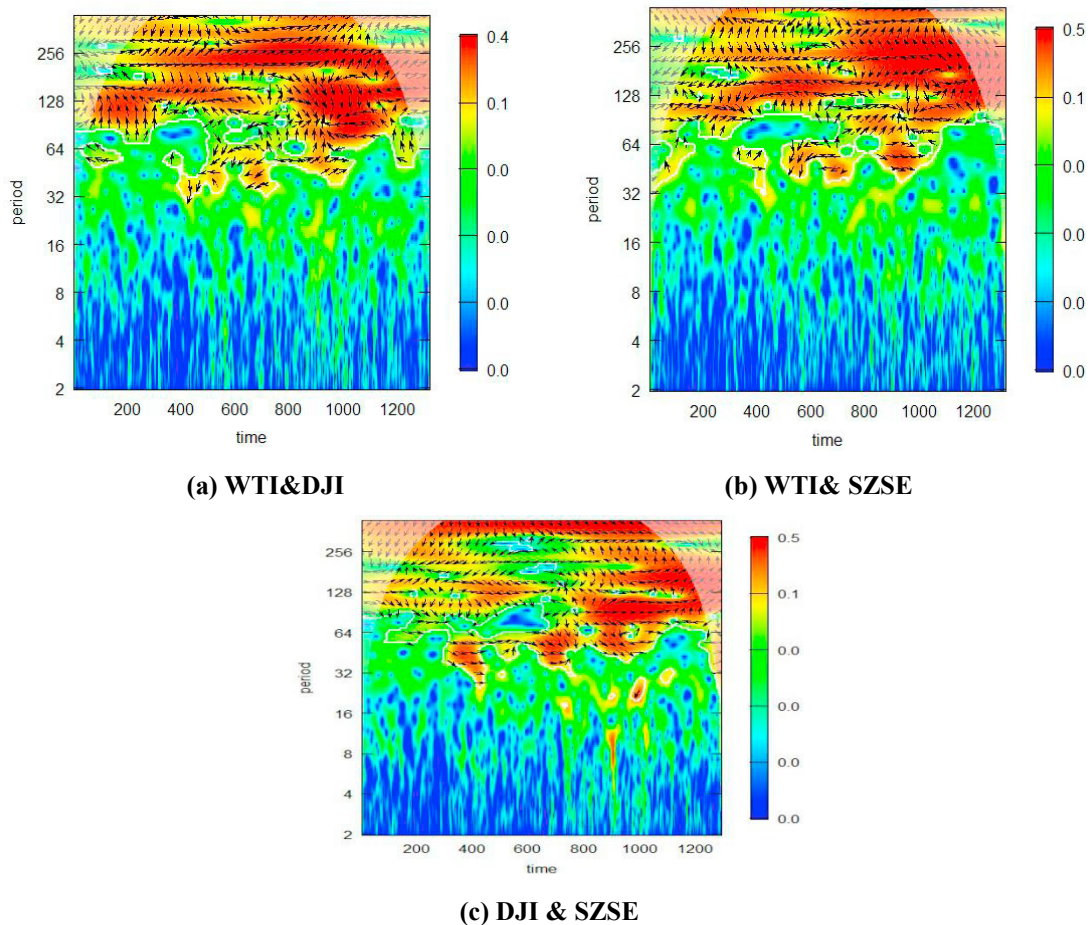


Fig. 2. Cross-wavelet transformation plots

We try a time-frequency view of the spectrum plot to check the frequency of internal changes as Figure 3. It shows the percentage of the time that a given frequency is present in a time series. The spectrum plot is the frequency stability status histogram. As the signal develops, the longer a specific frequency lasts in the signal, the higher its time sustains, so the brighter or hotter its color is on the screen. We use the persistence spectrum to identify signals hidden in these three time-series.

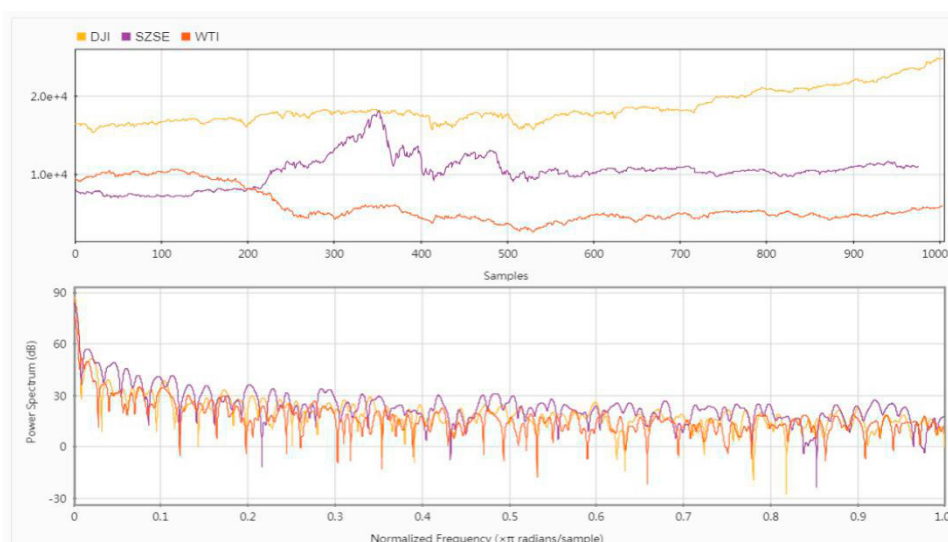


Fig. 3. Spectrum of time series

The spectrum of these three series indicates the existence of volatility clustering. Figure 3 shows that these three time series share similar signal periods and signal directions; therefore, if there is a downward impact on the WTI, there will be an internal downward crash factor in the DJI. The corresponding hedging strategy will be sold out or risk diversion applied.

4. Conclusion

This paper presents a wavelet transformation method to examine the multilevel pattern of co-movement with WTI oil market and stock markets. Time-varying dynamic correlation among the markets at different levels have been explored through wavelet coherence analysis. Significant long duration co-movement among different markets have been detected. The results show the multi-level chaotic nature of the financial market. This provides theoretical support for using discrete wavelet analysis. The chaotic nature of co-movement can be seen from the cross-wavelet transformation plots, which infers that a stable mode exists at level 128 or above. We use DWT to separate stable chaotic levels; however, DWT can only give the coefficients and have no ability to predict. Thus, we constructed spectrum plot to predict the key frequency for stable levels. To sum up, the co-movement mode among the three markets can be separated by DWT and prediction can be made using spectrum plot. The findings of the co-movement analysis can be effectively used for asset allocation and risk hedging across time periods.

Limitations to this research can be addressed as two points. Wavelet coefficients cannot identify the specific value of shock, only have the ability of direction display, so we can only infer there is co-movement between different markets, but we cannot get the degree of co-movement. Meanwhile, usage predictive modeling results in highly accurate forecasts, multivariate predictive models based on technical indicators should be investigated.

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